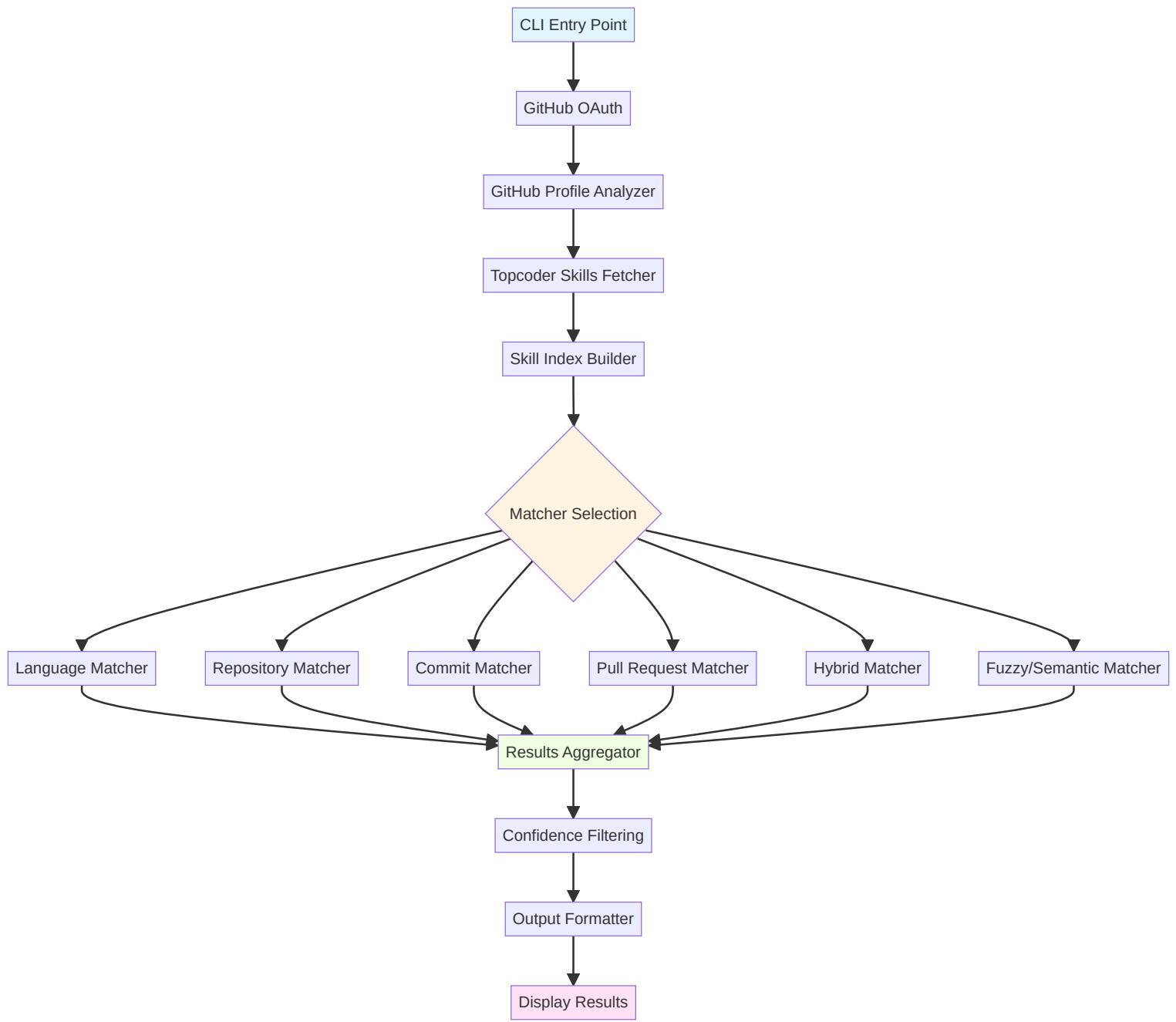


Architecture Documentation

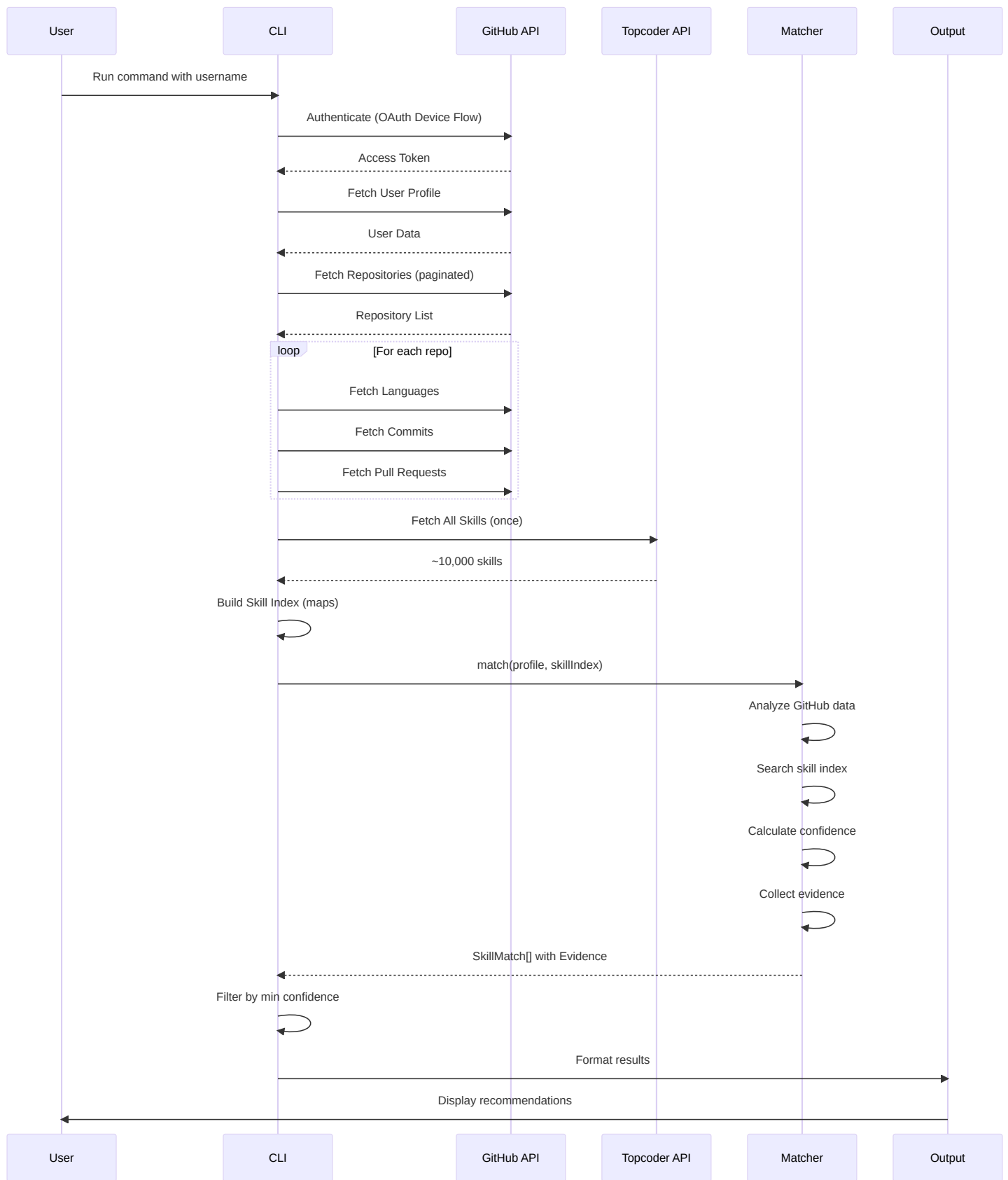
System Overview

The Topcoder Skills Recommender is a CLI tool that analyzes GitHub user profiles and recommends Topcoder standardized skills based on deep analysis of repositories, commits, pull requests, and code contributions.

High-Level Architecture

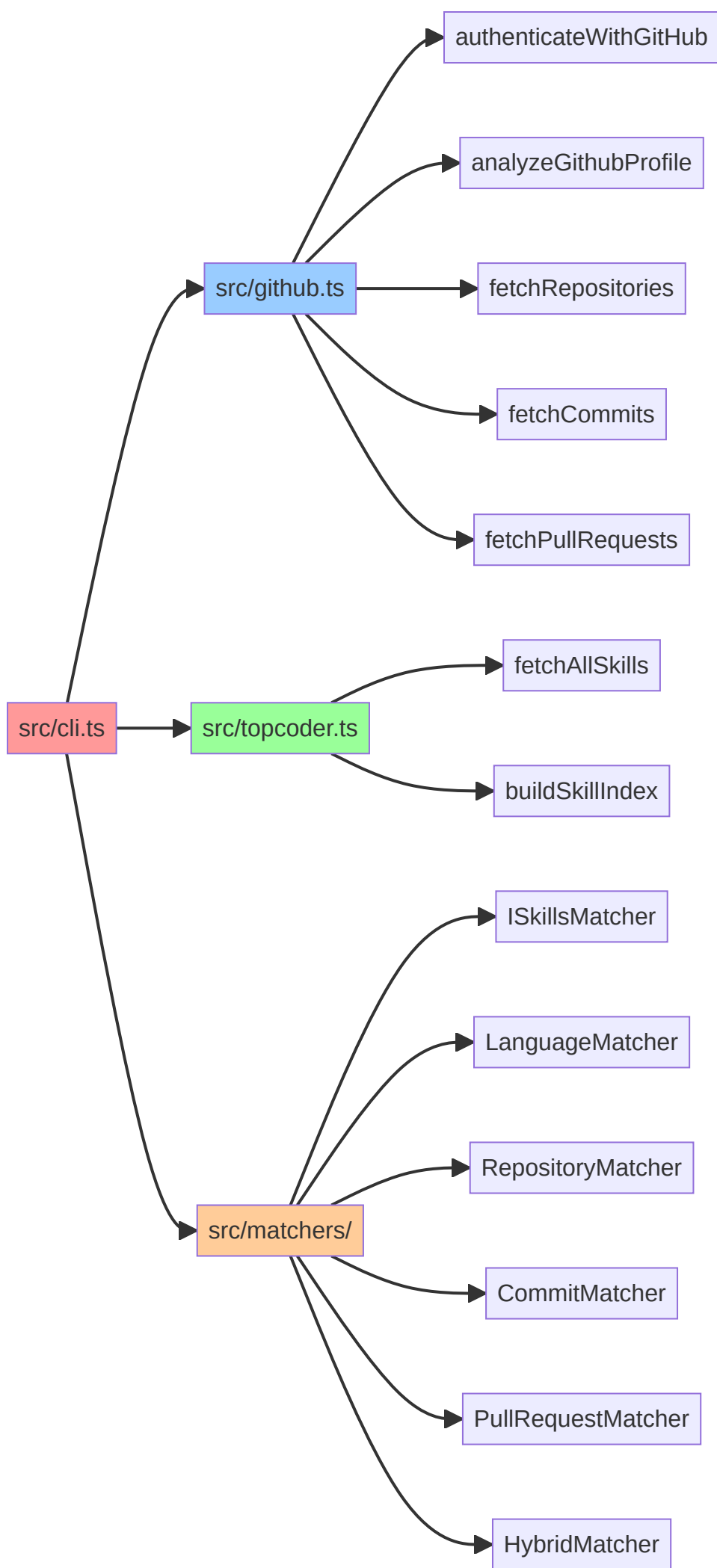


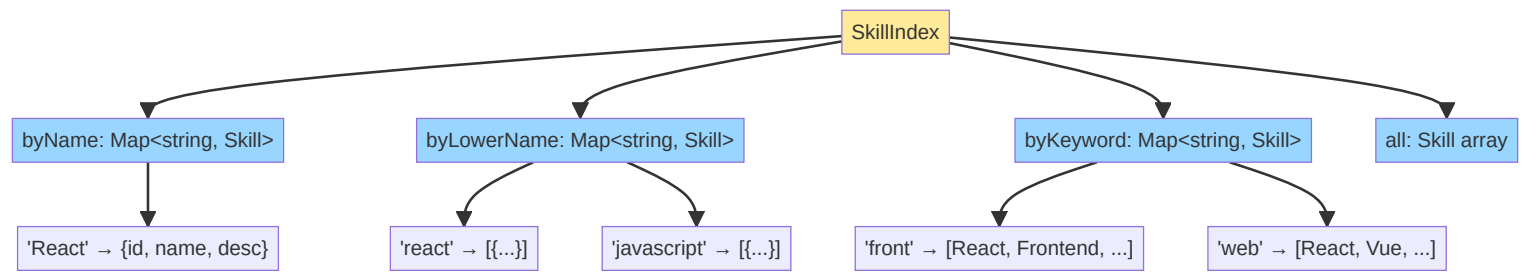
Data Flow



Component Architecture

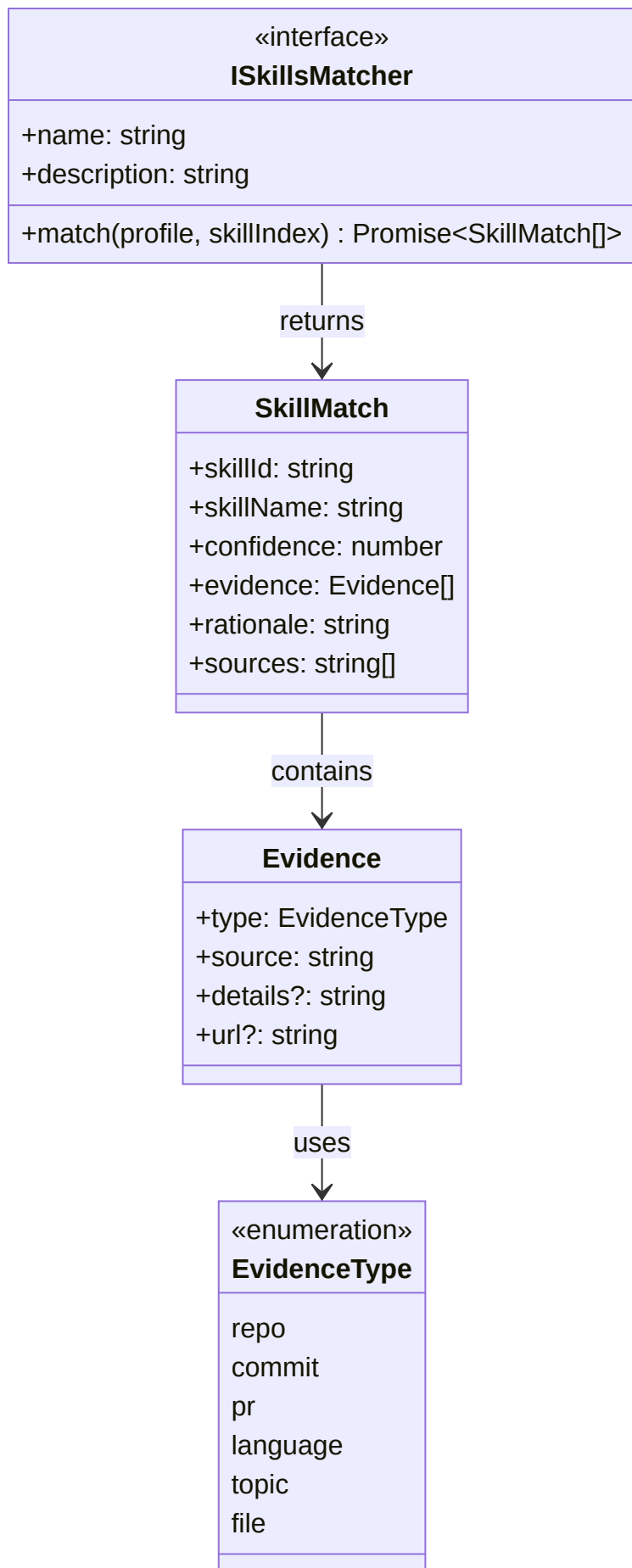
1. Core Components





Index Building Process: 1. Exact name: `"React"` → Direct $O(1)$ lookup 2. Case-insensitive: `"react"` → Array of matches 3. Keywords: Split skill name and description into words → Map word to skills 4. Enables multi-strategy matching without repeated API calls

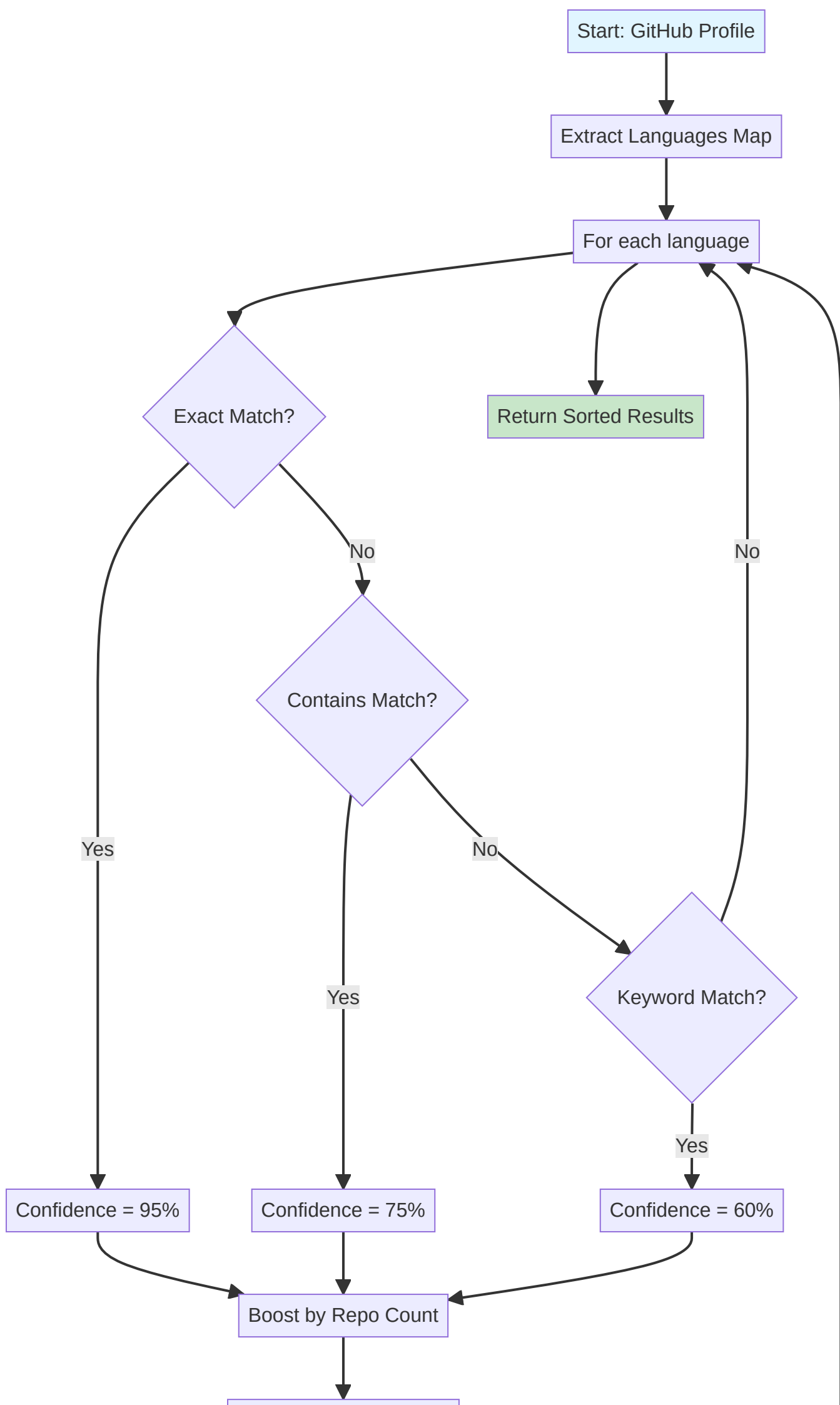
3. Matcher Interface

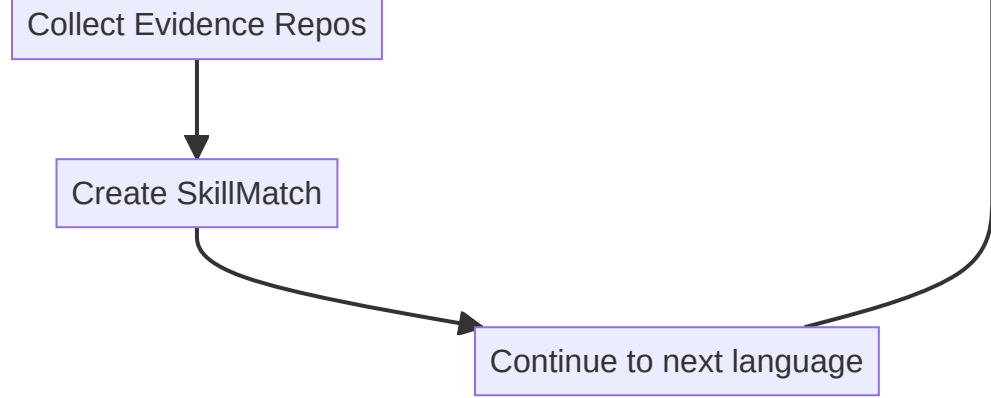


Key Interfaces: - **ISkillsMatcher**: Contract all matchers must implement - **SkillMatch**: Skill recommendation with confidence score - **Evidence**: Proof from GitHub (repos, commits, PRs) - **EvidenceType**: Classification of evidence source

Matcher Algorithms

1. Language Matcher (Most Reliable)

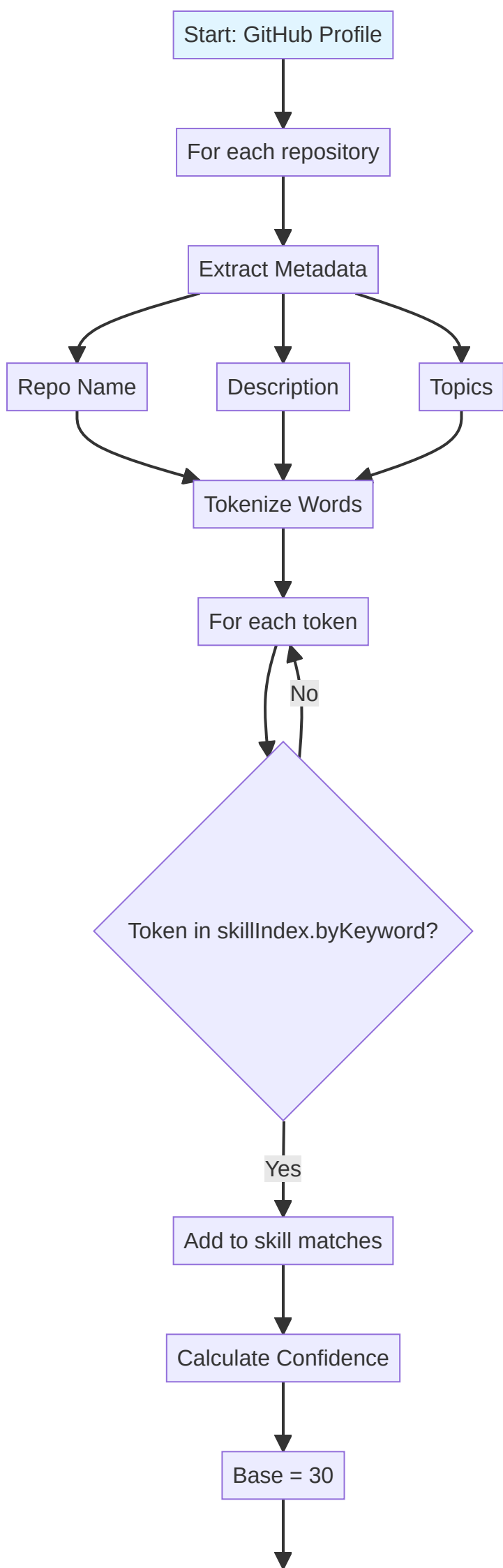


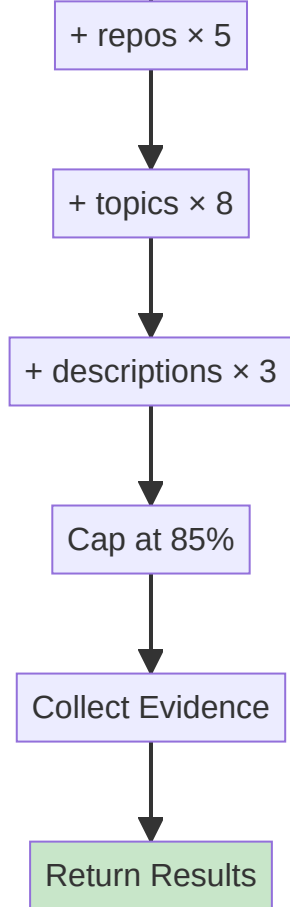


Algorithm:

1. Extract: `profile.languages` → `Map<string, number>`
2. For each language:
 - a. Exact match: `"JavaScript" === skill.name` → 95%
 - b. Partial: `"JavaScript" in skill.name` → 75%
 - c. Related: skill keywords contain `"javascript"` → 60%
3. Boost: +2% per repo (max +20%)
4. Evidence: List top 5 repos using this language
5. Return: Sorted by confidence (desc)

2. Repository Matcher

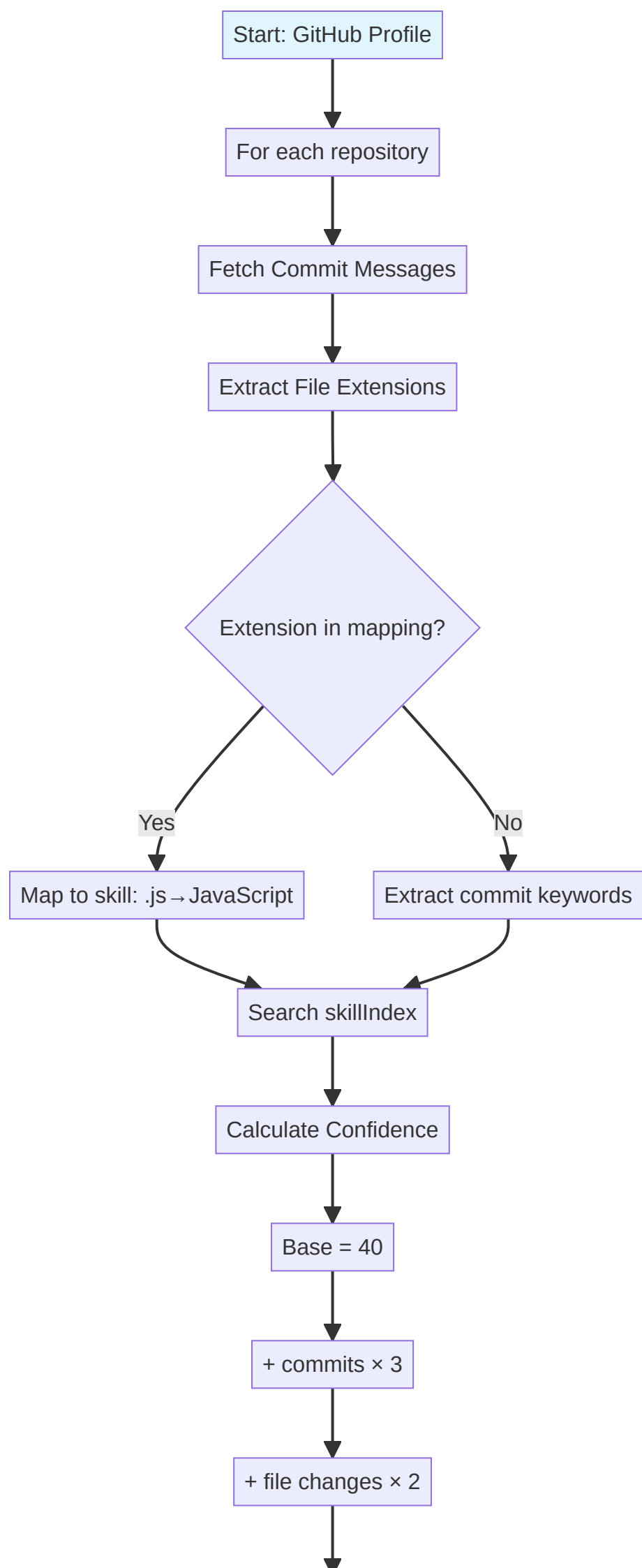


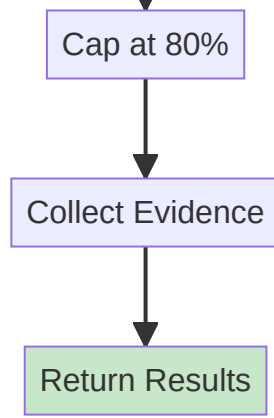


Confidence Formula:

```
confidence = min(30 + (matching_repos × 5) + (topics × 8) + (descriptions × 3), 85)
```

3. Commit Matcher

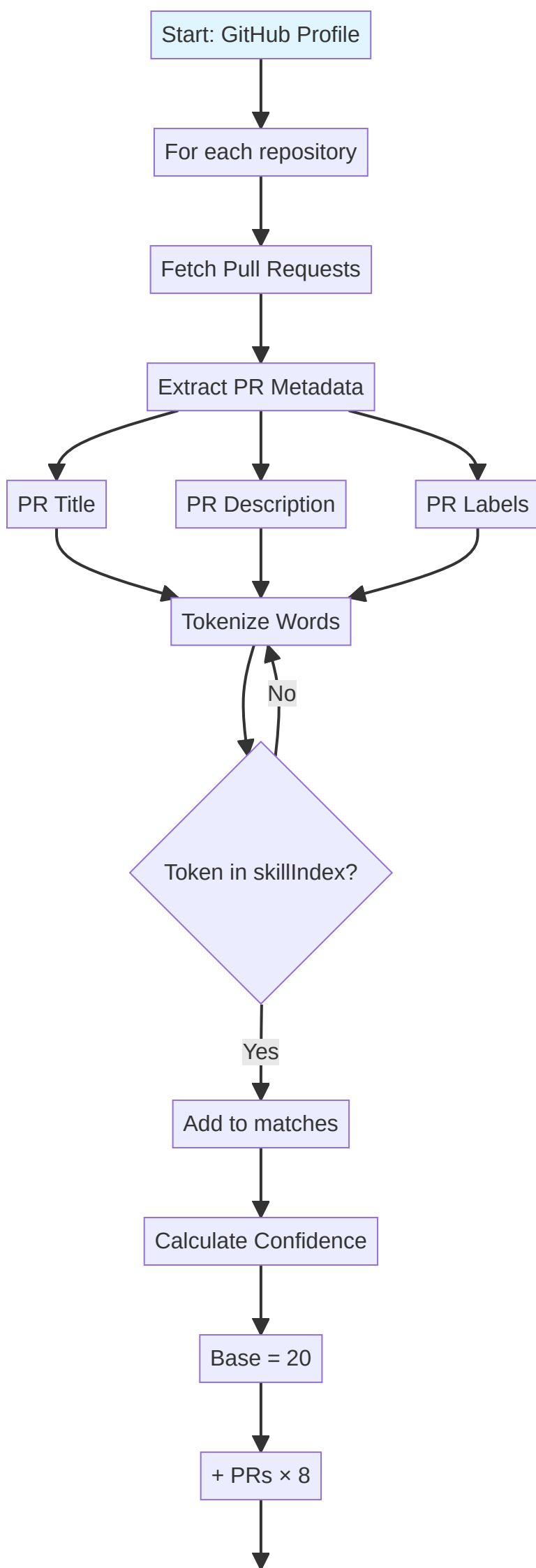


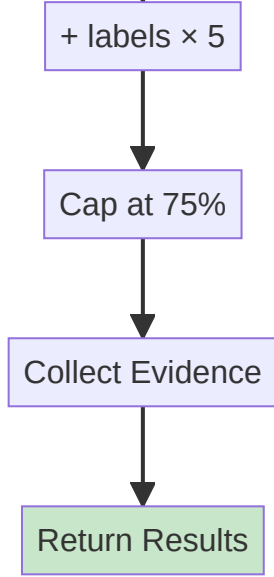


File Extension Mapping:

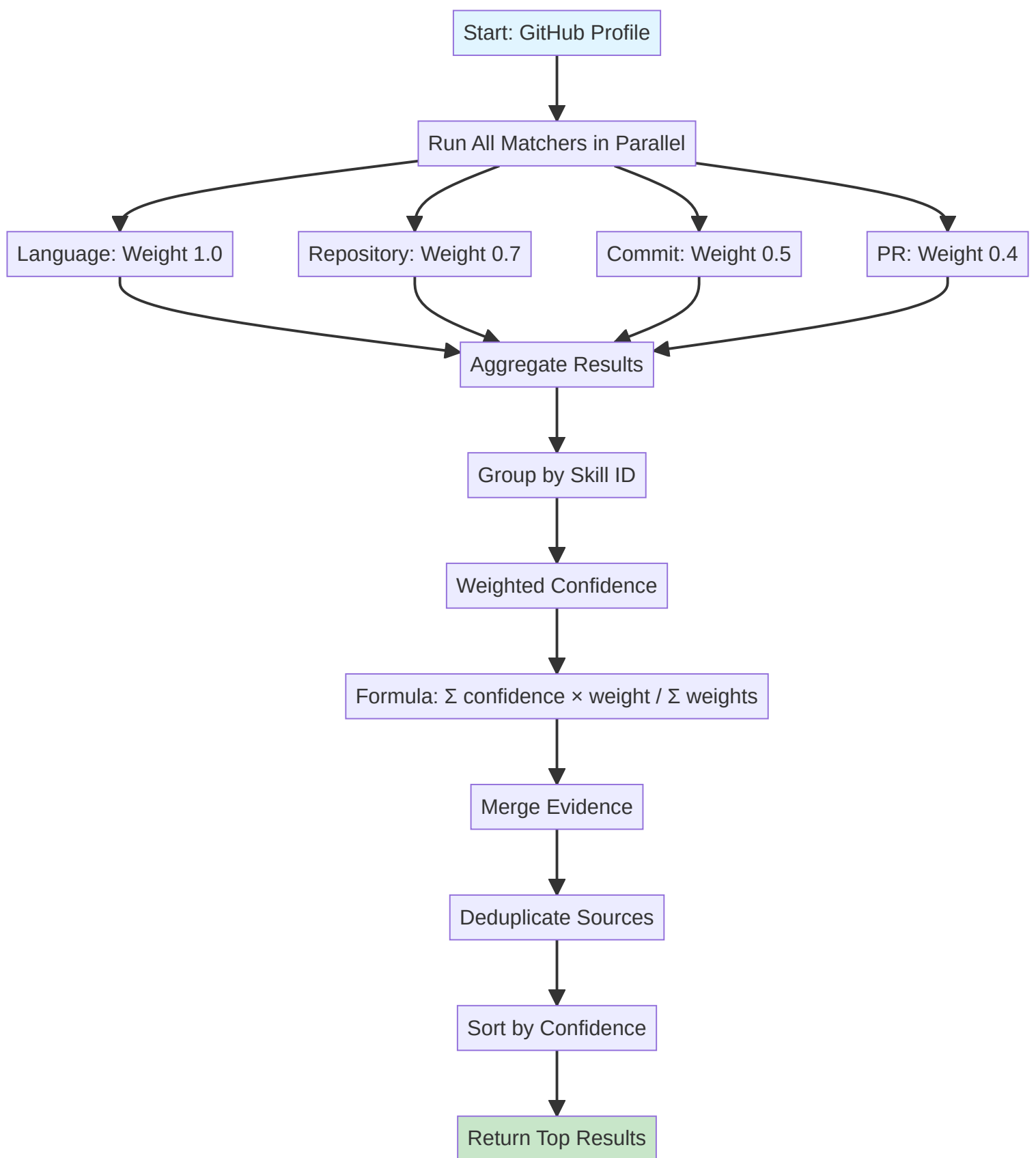
```
' .js' | '.jsx' → 'JavaScript'  
' .ts' | '.tsx' → 'TypeScript'  
' .py' → 'Python'  
' .java' → 'Java'  
' .go' → 'Go'  
' .rs' → 'Rust'  
// ... etc
```

4. Pull Request Matcher



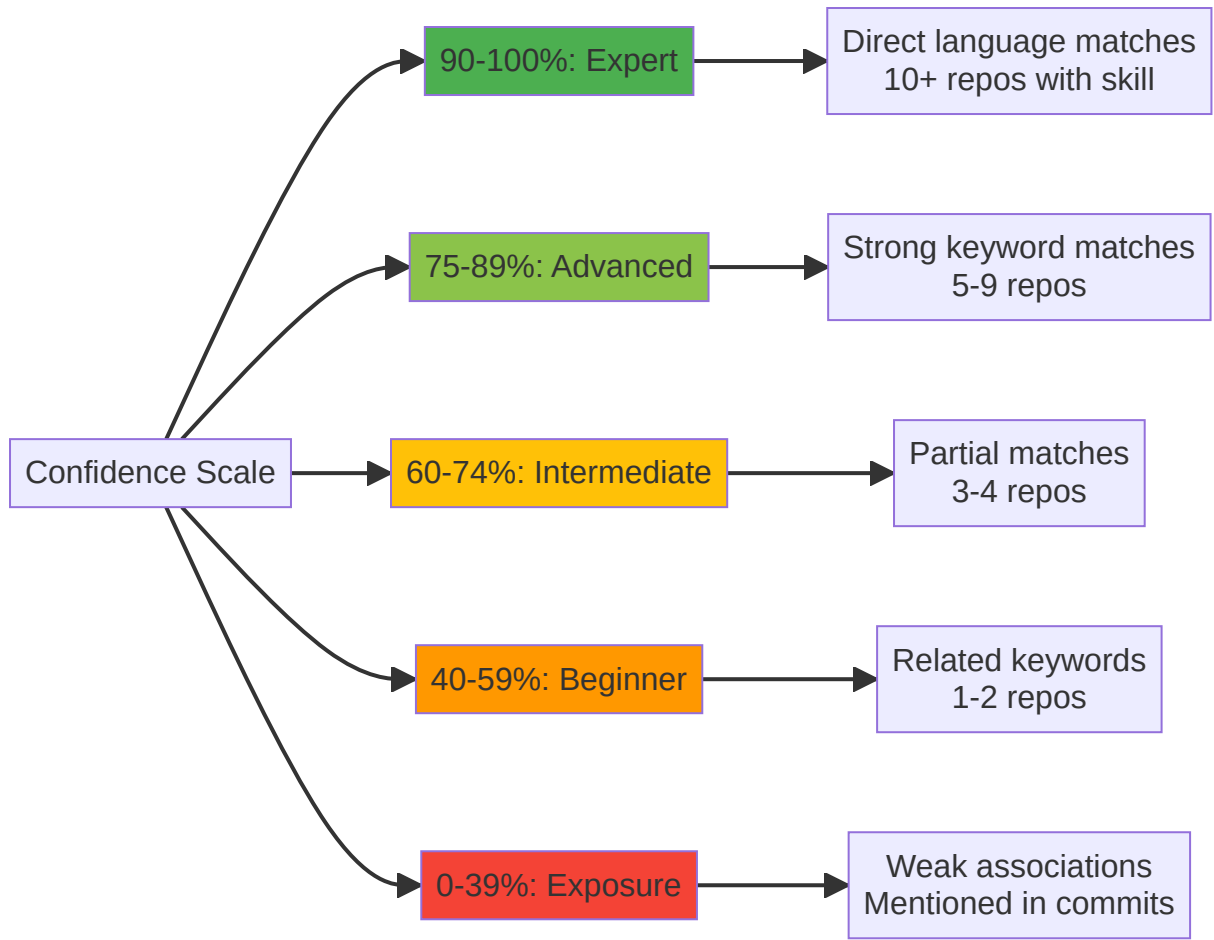


5. Hybrid Matcher (Recommended)



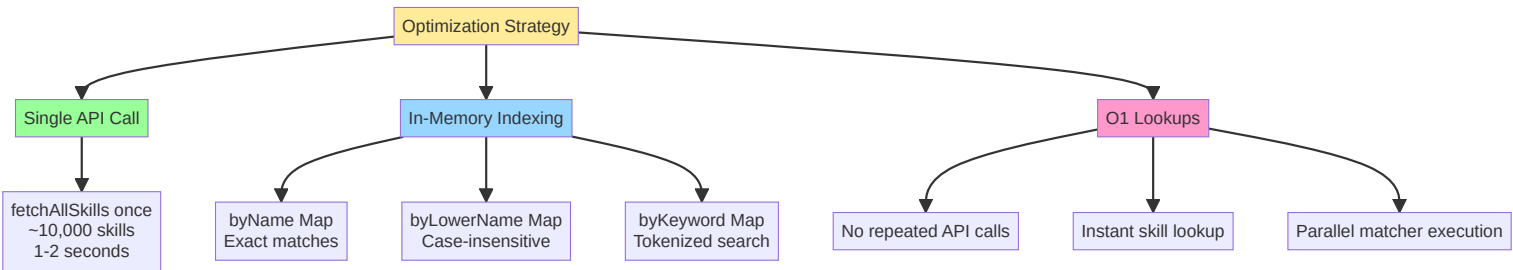
Weighted Combination:

```
For each skill found by multiple matchers:  
weighted_confidence = (  
    language_confidence × 1.0 +  
    repo_confidence × 0.7 +  
    commit_confidence × 0.5 +  
    pr_confidence × 0.4  
) / total_weights_used
```



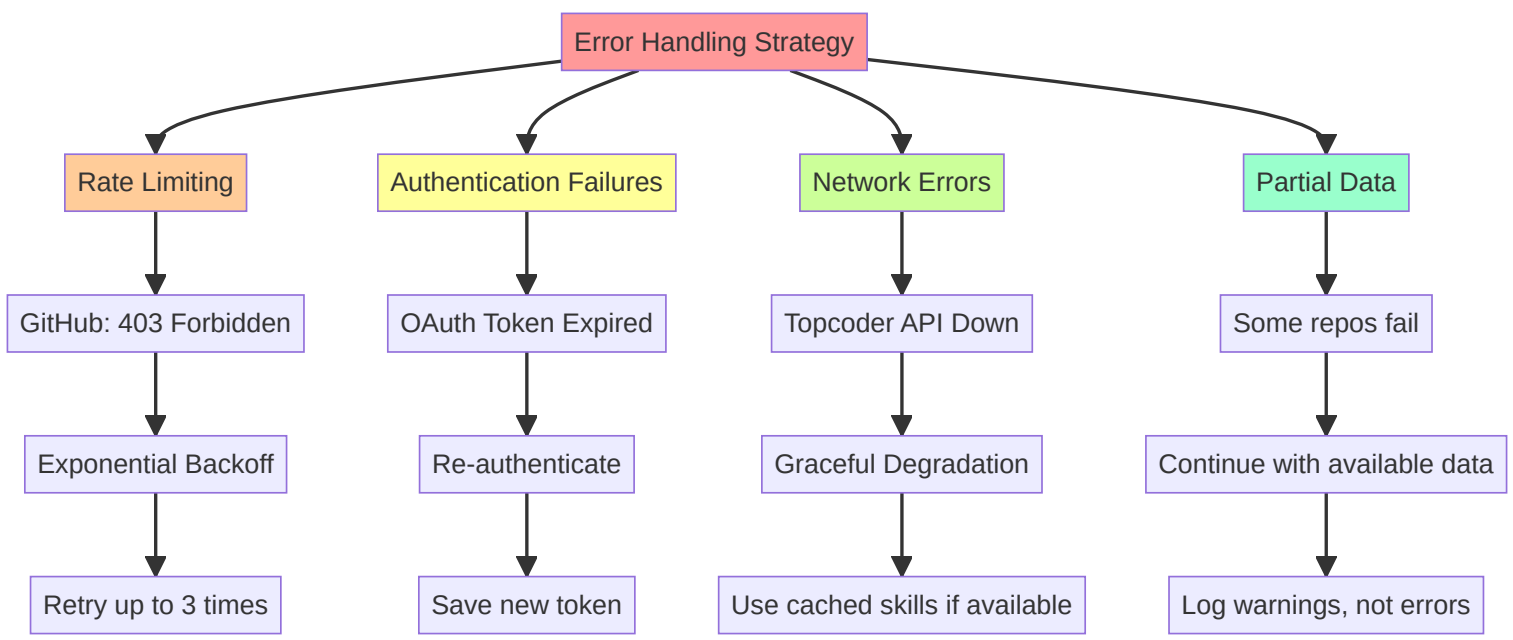
Performance Optimization

Skill Index Strategy



Before vs After: - **Before:** 100+ API calls per user (fuzzy match, semantic search for each language) - **After:** 1 API call per session (fetch all skills once, match locally) - **Speed:** 10-30 seconds → 2-5 seconds - **Reliability:** Network failures → None (local matching)

Error Handling



Output Formats

Text Format

🎯 Topcoder Skill Recommendations for octocat

1. JavaScript (98% confidence)

Evidence:

- Language: frontend-app (42 contributions)
- Language: backend-api (38 contributions)
- Repository: js-toolkit (topic: javascript)

Found by: Language Matcher, Repository Matcher

2. React (95% confidence)

Evidence:

- Language: react-dashboard (56 contributions)
- Repository: react-components (description: "React UI library")

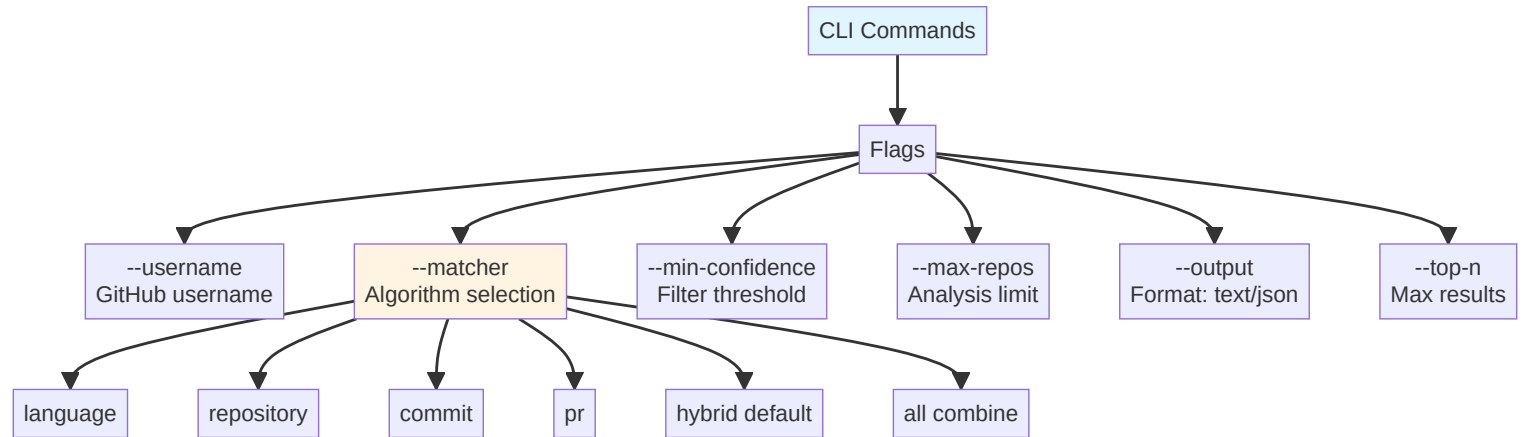
Found by: Language Matcher, Hybrid Matcher

JSON Format

```
{
  "username": "octocat",
  "matcher": "hybrid",
  "minConfidence": 50,
  "recommendations": [
    {
      "skillId": "abc123",
      "skillName": "JavaScript",
      "confidence": 98,
      "evidence": [
        {
          "type": "language",
          "source": "frontend-app",
          "details": "42 contributions",
          "url": "https://github.com/octocat/frontend-app"
        }
      ]
    },
    {
      "skillId": "def456",
      "skillName": "React",
      "confidence": 95,
      "evidence": [
        {
          "type": "language",
          "source": "react-dashboard",
          "details": "56 contributions",
          "url": "https://github.com/octocat/react-dashboard"
        },
        {
          "type": "repository",
          "source": "react-components",
          "details": "React UI library",
          "url": "https://github.com/octocat/react-components"
        }
      ]
    }
  ],
  "rationale": "Exact language match with high frequency",
  "sources": ["Language Matcher", "Repository Matcher"]
}
```

```
}
```

CLI Interface



Example Usage:

```
# Default (hybrid matcher, min 50% confidence)
npm start -- --username octocat

# Language-only matching
npm start -- --username octocat --matcher language

# High confidence only
npm start -- --username octocat --min-confidence 75

# JSON output for API integration
npm start -- --username octocat --output json > results.json

# Run all matchers and combine
npm start -- --username octocat --matcher all --top-n 20
```

Security & Privacy

- **OAuth Token Storage:** Stored in `.github-token` file (gitignored)
- **No Data Persistence:** GitHub profile data not stored, processed in-memory only
- **API Keys:** Environment variables for any optional AI providers
- **Rate Limits:** Respects GitHub API limits (5000/hour authenticated)

Extensibility

The architecture supports easy addition of new matchers - see [PLUGIN_GUIDE.md](#) for step-by-step instructions.